LEVEL UP ACADEMY

Class 10 | CBSE | Science

Pre-Board Mock 2 (CBSE pattern)

Chapter / Topic: Full syllabus (Chapters 1–13)

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. This paper has 39 questions in 5 sections (A–E). All questions are compulsory. Internal choice is given in some questions.
2. Section A: Q1–Q20, 1 mark each (MCQs and Assertion–Reason). Section B: Q21–Q26, 2 marks each. Section C: Q27–Q33, 3 marks each.
3. Section D: Q34–Q36, 5 marks each. Section E: Q37–Q39, case-based, 4 marks each.
4. Write balanced equations, show formulae and units in numericals, and draw neat labelled diagrams wherever required.
5. This mock follows the reported 2026-27 pattern. Check the official CBSE sample paper for any changes.

SECTION A

20 marks

Q1–Q16: choose the correct option. Q17–Q20: Assertion–Reason. For these, choose (a) both A and R true and R explains A; (b) both true but R does not explain A; (c) A true, R false; (d) A false, R true.

Q1. Which of the following is a double displacement reaction? **[1]**

(a) $2Mg + O_2 \rightarrow 2MgO$ (b) $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$
(c) $CuO + H_2 \rightarrow Cu + H_2O$ (d) $2H_2O \rightarrow 2H_2 + O_2$

Q2. A student adds dilute hydrochloric acid to zinc granules. What will the student observe? **[1]**

(a) The solution turns blue
(b) A white precipitate forms
(c) No change
(d) Gas bubbles form around the zinc granules

Q3. Which of these is a weak acid? **[1]**

(a) HCl (b) CH_3COOH
(c) H_2SO_4 (d) HNO_3

Q4. An element X forms a basic oxide X_2O and reacts vigorously with cold water. X is most likely: **[1]**

(a) carbon (b) sulphur
(c) sodium (d) aluminium

Q5. The molecular formula of an alcohol with three carbon atoms is: **[1]**

(a) C_3H_6O (b) C_2H_5OH
(c) C_3H_8 (d) C_3H_7OH

Q6. Which of these is used to remove permanent hardness of water? **[1]**

(a) Baking soda (b) Bleaching powder
(c) Plaster of Paris (d) Washing soda

Q7. Which enzyme in pancreatic juice digests proteins? **[1]**

(a) Amylase (b) Lipase
(c) Trypsin (d) Pepsin

Q8. The products of anaerobic respiration in yeast are: **[1]**

(a) lactic acid and energy
(b) ethanol, carbon dioxide and energy
(c) carbon dioxide, water and energy

(d) glucose and oxygen

Q9. Which part of the brain controls involuntary actions such as blood pressure, salivation and vomiting? **[1]**

(a) Medulla (b) Cerebrum
(c) Cerebellum (d) Hypothalamus

Q10. In human females, fertilisation takes place in the: **[1]**

(a) ovary (b) oviduct
(c) uterus (d) vagina

Q11. In Mendel's dihybrid cross, how many different phenotypes appear in the F_2 generation? **[1]**

(a) 2 (b) 3
(c) 4 (d) 9

Q12. Which of these is a producer in a pond ecosystem? **[1]**

(a) Fish (b) Frog
(c) Algae (d) Crab

Q13. An object is placed between the pole and the focus of a concave mirror. The image formed is: **[1]**

(a) real and inverted (b) virtual and diminished
(c) real and of the same size (d) virtual, erect and enlarged

Q14. The power of a concave lens of focal length 25 cm is: **[1]**

(a) −4 D (b) −0.25 D
(c) +0.25 D (d) +4 D

Q15. Three resistors of 6 Ω each are connected in parallel. The equivalent resistance is: **[1]**

(a) 2 Ω (b) 3 Ω
(c) 6 Ω (d) 18 Ω

Q16. Which of the following is caused by the scattering of light? **[1]**

(a) Blue colour of the sky (b) Twinkling of stars
(c) Advance sunrise (d) Formation of a rainbow

Q17. **Assertion (A):** Blue copper sulphate crystals turn white on heating.
Reason (R): Heating removes the water of crystallisation. **[1]**

Q18. **Assertion (A):** Plants exchange gases through the stomata.
Reason (R): Stomata are usually closed at night to reduce water loss. **[1]**

Q19. **Assertion (A):** A fuse wire has a low melting point.
Reason (R): The fuse wire must melt and break the circuit when an unduly high current flows. **[1]**

Q20. **Assertion (A):** Ozone in the upper atmosphere is essential for life on Earth.
Reason (R): Ozone is a deadly poison at ground level. **[1]**

SECTION B

12 marks

Q21. Why are decomposition reactions called the opposite of combination reactions? Give one balanced equation for each type. **[2]**

Q22. What is meant by water of crystallisation? Give the name and formula of one salt that contains it. **[2]**

OR

Why does an aqueous solution of an acid conduct electricity?

Q23. Name the components of the transport system in human beings and give the function of each. **[2]**

Q24. What is regeneration? Why can't complex organisms reproduce by regeneration? **[2]**

OR

What is a pollen tube? What is its role in fertilisation?

Q25. The speed of light in glass is 2×10^8 m/s. Find the refractive index of glass. (Speed of light in vacuum = 3×10^8 m/s.) **[2]**

Q26. Why are electric bulbs filled with chemically inactive gases like nitrogen and argon? Why does the tungsten filament not melt? **[2]**

OR

Three 60 W bulbs are used for 4 hours every day. How many units (kWh) of energy do they use in 30 days?

SECTION C

21 marks

Q27. Write balanced chemical equations for the following: **[3]**

(i) Calcium carbonate is heated. **[1]**

(ii) Methane burns in oxygen. **[1]**

(iii) Sodium hydroxide reacts with dilute sulphuric acid. **[1]**

Q28. Answer the following about the hormone that controls the metabolism of carbohydrates, fats and proteins: **[3]**

(i) Name the hormone. **[1]**

(ii) Which gland secretes it? **[1]**

(iii) Which mineral is needed to make it, and what disease does its deficiency cause? **[1]**

Q29. How would you tell ethanol and ethanoic acid apart? Describe two tests, and write the equation for one of them. **[3]**

Q30. Explain the following movements in plants, with one example each: (i) phototropism, (ii) geotropism, (iii) chemotropism. **[3]**

Q31. How is the small intestine designed to absorb digested food? **[3]**

OR

Give the role of each of the following in the human body: (i) lymph, (ii) platelets, (iii) valves in the heart.

Q32. An object is placed 10 cm in front of a concave mirror of focal length 15 cm. Find the position, nature and magnification of the image. **[3]**

OR

A 3 cm tall object is placed 18 cm from a convex lens of focal length 12 cm. Find the position, size and nature of the image.

Q33. A copper wire is 50 m long and has a cross-sectional area of 1 mm^2 (1×10^{-6} m^2). The resistivity of copper is 1.7×10^{-8} Ω m. **[3]**

(i) What is the SI unit of resistivity? **[1]**

(ii) Find the resistance of the wire. **[1]**

(iii) What would the resistance be if the length were halved? **[1]**

SECTION D

15 marks

Q34. Answer the following: **[5]**

(i) With one balanced equation each, show how a metal and a non-metal react with oxygen. What is the nature of each oxide? **[3]**

(ii) Why do silver articles turn black, and copper vessels turn green, over time? **[2]**

OR

(a) What is a functional group? Name the functional group, with its formula, in ethanol, ethanal, propanone and ethanoic acid. [3]
(b) Write the name and formula of the second member of the alkene series. [2]

Q35. Answer the following: **[5]**

(i) Describe how urine is formed in a nephron. **[3]**

(ii) What decides the amount of urine produced? **[2]**

OR

(a) Give two differences between asexual and sexual reproduction. [2]
(b) After fertilisation in a flower, explain how the seed and the fruit are formed. [3]

Q36. Answer the following: **[5]**

(i) What is meant by the power of accommodation of the eye? **[1]**

(ii) A student sitting at the back of the class cannot read the board clearly, but can read a book. Name the defect, give two causes, and explain how it is corrected. **[3]**

(iii) Why can a normal eye not see clearly an object placed closer than 25 cm? **[1]**

OR

(a) Give two properties of magnetic field lines. [2]
(b) How does the magnetic field due to a current-carrying circular loop change if (i) the current is increased, (ii) the number of turns is increased, (iii) the distance from the wire is increased? [3]

SECTION E

12 marks

Read each passage and answer the questions that follow.

Q37. In an experiment, three test tubes each held an iron nail. In tube A, the nail was in tap water with air above it. In tube B, the nail was in boiled distilled water covered with a layer of oil. In tube C, the nail was in dry air with anhydrous calcium chloride (which absorbs moisture). The tubes were left for a few days. **[4]**

(i) In which tube did the nail rust? **[1]**

(ii) Why was the water in tube B boiled and covered with oil? **[1]**

(iii) What does this experiment show about rusting? Give one way to prevent it. **[2]**

Q38. The force that blood exerts against the wall of a blood vessel is called blood pressure. The pressure during ventricular contraction (systolic) is normally about 120 mm of Hg, and during relaxation (diastolic) about 80 mm of Hg. Blood pressure is measured with an instrument called a sphygmomanometer. High blood pressure, or hypertension, is caused by the narrowing (constriction) of arterioles. **[4]**

(i) What is the normal blood pressure of a human being? **[1]**

(ii) Name the instrument used to measure blood pressure. **[1]**

(iii) What causes hypertension, and why is it dangerous? **[2]**

Q39. A household uses a 1000 W geyser for 1 hour a day, four 10 W LED bulbs for 6 hours a day, and a 100 W television for 3 hours a day. Electricity costs ₹6 per unit (kWh). **[4]**

(i) How much energy does the geyser use per day? **[1]**

(ii) What is the total energy used per day? **[1]**

(iii) Find the electricity bill for 30 days. Why are these appliances connected in parallel? **[2]**

~ All the best for your boards! ~